

Math 2551 Worksheet 18: Triple Integrals

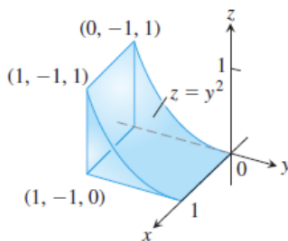
1. Evaluate the triple iterated integral

$$\int_{-1}^1 \int_0^4 \int_0^1 z^3 - 4x^2y \, dz \, dy \, dx.$$

When you evaluate the innermost integral, you should treat both x and y as constants and take the antiderivative with respect to z .

What is the region of integration for this integral in $\mathbb{R}^3$?

2. Set up a triple iterated integral for $\iiint_E z \, dV$, where E is the solid tetrahedron in the first octant bounded above by $x + y + z = 1$. It may be helpful to make a sketch of the solid.
3. Set up integrals that would calculate the volume of the region below, using the specified orders of integration.



- (a) $dy \, dz \, dx$ (b) $dy \, dx \, dz$ (c) $dx \, dy \, dz$ (d) $dx \, dz \, dy$ (e) $dz \, dx \, dy$

4. Let D be the region bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 2y$, i.e.

$$D = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq z \leq 2y\}.$$

Write triple iterated integrals in the orders $dz \, dy \, dx$ and $dx \, dz \, dy$ that give the volume of D . Can you write a single triple iterated integral for this volume using any other orders of integration?